

DH 604

FINAL REPORT PRESENTATION



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PROBLEM STATEMENT REFINED VERSION

Problem Statement (Modified)

To implement an architecture of a guided multi-turn conversational system capable of deeply personalized interaction by utilising long-term user history, persona attributes, and dynamic behavioral records.

Traditional Supervised Fine-Tuning (SFT) or Unsupervised approaches are inherently limited for this task. They optimize for the immediate next token, effectively treating conversation as a static pattern-matching exercise. However, human conversation is stochastic—filled with irregularities, randomness, and delayed dependencies that standard tuning fails to capture. Our architecture has to be trained to:

3. Analyze Context (S_t): Dynamically encode user persona, records, and long-term history (H_t) to understand the current state.
4. Optimize Policy (π): Learn a conversational policy that balances immediate engagement $R(\text{now})$ with long-term strategic goals $R(\text{final})$.

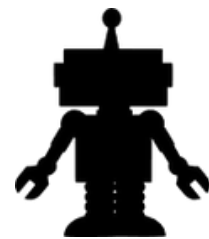
Given the dynamic state S_t , which is derived from the historical information H_t and current attributes A_t , we want to design a reasoning architecture P that produces a sequence of context representations C_t that take a non myopic approach which is, rather than focusing only on minimizing prediction error at the current time step t , the primary objective of model P is to maximize the Cumulative Future Reward (G), that not only gives a good immediate response but also takes the conversation through the optimal policy.]

Reasoning-Based Task Scheduler: At time $t=T$, this component functions as a fine-tuned decision-making agent rather than a simple retrieval system. Instead of passively fetching items from a knowledge graph, it utilizes the context C_t to actively reason about the user's needs by synthesizing current circumstances, historical behavioral patterns, and persona. The scheduler generates specific tasks—such as recommendations or follow-ups—and determines the underlying rationale for *why* a specific task is optimal at that moment

Example



My doctor told me last week that I have high blood pressure, around 155 over 95.



- Seek hypertension evaluation: systolic BP ≥ 140 mmHg
- Arrange a comprehensive cardiovascular risk assessment (e.g., lipid panel, fasting glucose, ECG) to quantify additional risk factors associated with the elevated blood pressure.
- Schedule a personalized lifestyle counseling session (dietary advice, exercise plan, weight-management strategies) to support blood-pressure control.



I was recently diagnosed with type 2 diabetes. My HbA1c came back at 8.3 percent.



- Schedule endocrinology review: HbA1c above target ($>7\%$) for diabetic patient
- Consult your doctor: HbA1c above 6.5% — diabetes screening recommended

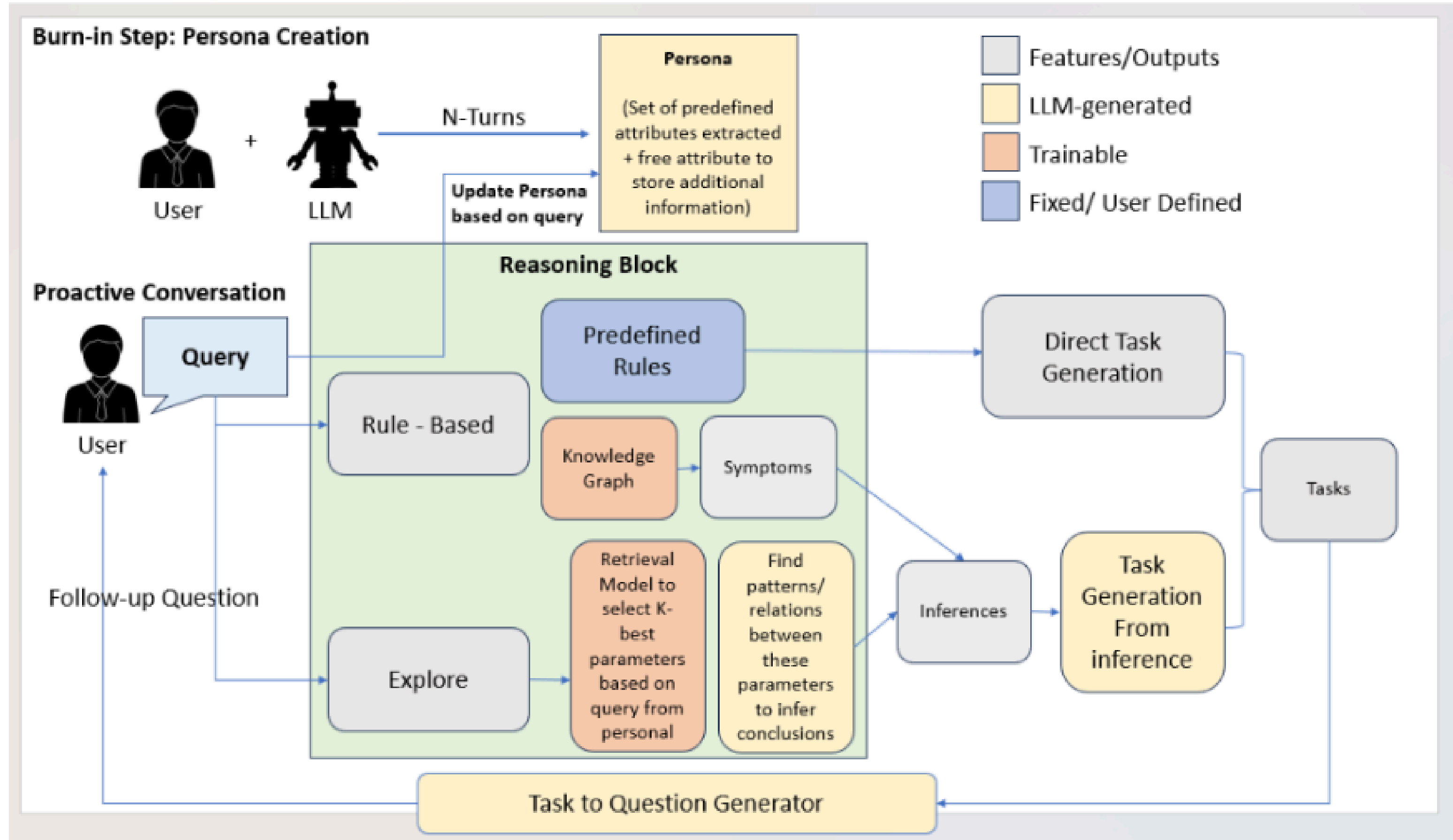


I usually sleep around 5 hours a night because of work stress. I feel exhausted all the time.



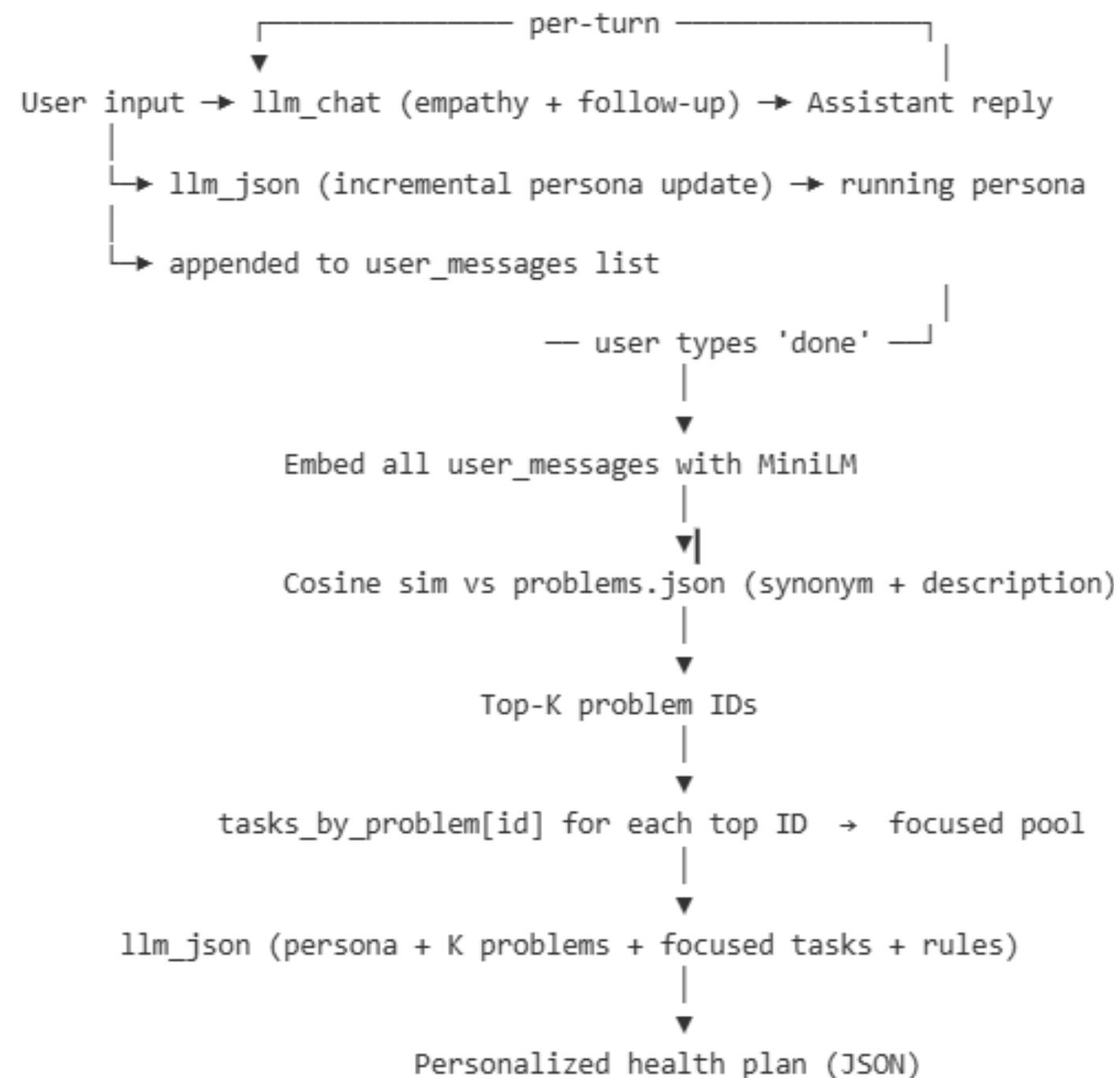
- Try to get more sleep to reduce blood pressure
- Schedule a home sleep apnea screening test to evaluate for obstructive sleep apnea, which can worsen hypertension, diabetes control, and sleep quality

Proposed Architecture



BASELINE VERSION 1 - SEMANTIC SEARCH

Main Architecture



Problem Statement

Given an unstructured, multi-turn natural-language description of a user's symptoms, the system must:

- Build a structured user persona aligned with an Ayurvedic schema.
- Identify the user's top conditions from a curated problem dataset.
- Recommend a personalized set of Ayurvedic and lifestyle tasks for the detected conditions.

Goal

Build an intelligent diagnosis system that can :

- Turn by turn conversation via LLM
- Updating the user persona along with storing user symptoms
- Embed problem symptoms and user symptoms and perform cosine similarity
- Gives top 3 diseases along with task list

BASELINE VERSION 1 - SEMANTIC SEARCH

problems.json

```
{
  "id": "PSY_002",
  "primary_term": "Insomnia",
  "iks_term": "Anidra",
  "category": "Mental Health",
  "description": "Difficulty falling asleep, staying asleep, or waking up too often",
  "synonyms": [
    "I can't sleep at night",
    "trouble falling asleep",
    "I keep waking up at night",
    "insomnia",
    "sleepless nights",
    "I lie awake for hours",
    "poor sleep quality",
    "wake up tired",
    "disturbed sleep",
    "mind keeps running at bedtime"
  ],
  "relevant_attribute_paths": [
    "mental_emotional_health.sleep_quality",
    "mental_emotional_health.stress_level",
    "mental_emotional_health.emotional_dominance",
    "ayurvedic_constitution.gunas",
    "ayurvedic_constitution.vikriti",
    "lifestyle_factors.activity_level"
  ],
  "exploitation_rules": {
    "vata_pacification": "If user reports racing thoughts or restlessness, recommend grounding exercises.",
    "stress_link": "Cross-check with stress_level – high stress is the primary driver of insomnia."
  }
}
```

user_attributes.json

```
{
  "mental_emotional_health": {
    "description": "Psychological state (Manovaha Srotas)",
    "attributes": {
      "stress_level": {
        "name": "Stress Load",
        "type": "scale",
        "range": "1-10",
        "iks_term": "Chinta"
      },
      "sleep_quality": {
        "name": "Nidra",
        "type": "categorical",
        "values": ["Sound Sleep", "Insomnia (Anidra)", "Excessive Sleep"]
      },
      "emotional_dominance": {
        "name": "Predominant Emotion",
        "type": "list",
        "values": ["Anxiety (Chitta Udvega)", "Anger (Krodha)", "Grief (Shoka)", "Fear (Bhaya)", "Attachment (Moha)"]
      },
      "cognitive_function": {
        "name": "Focus & Memory",
        "type": "categorical",
        "values": ["Sharp", "Foggy/Confused", "Forgetful"]
      }
    }
  },
  "ayurvedic_constitution": {
    "description": "Individual's constitution (Prakriti)",
    "attributes": {
      "doshas": {
        "name": "Dosha Balance",
        "type": "scale",
        "range": "Vata/Pitta/Kapha",
        "iks_term": "Dosha"
      },
      "gunas": {
        "name": "Guna Balance",
        "type": "scale",
        "range": "Sattva/Rajas/Tamas",
        "iks_term": "Guna"
      },
      "vikriti": {
        "name": "Imbalance Type",
        "type": "list",
        "values": ["Vata", "Pitta", "Kapha", "Mixed"]
      }
    }
  },
  "lifestyle_factors": {
    "description": "Daily habits and routines",
    "attributes": {
      "activity_level": {
        "name": "Physical Activity",
        "type": "scale",
        "range": "Sedentary to Active",
        "iks_term": "Vyayama"
      },
      "dietary_habits": {
        "name": "Dietary Habits",
        "type": "list",
        "values": ["Balanced", "High Sugar/Fat", "Low Fiber"]
      },
      "sleep_schedule": {
        "name": "Sleep Schedule",
        "type": "list",
        "values": ["Regular", "Irregular", "Shift Work"]
      }
    }
  }
}
```

tasks.json

```
{
  "PSY_002": {
    "primary_term": "Insomnia",
    "iks_term": "Anidra",
    "task_list": [
      "Stick to a consistent sleep schedule, even on weekends.",
      "Drink warm nutmeg milk 30 minutes before bed.",
      "Keep the bedroom dark, cool, and quiet.",
      "Practice box breathing (4-4-4-4) before sleep.",
      "Massage the soles of the feet with oil.",
      "Avoid caffeine after 2 PM and alcohol at night.",
      "Practice 10 minutes of Yoga Nidra or Transcendental Meditation."
    ]
  }
}
```

6. Results

6.1 Summary

Metric	Score	Percentage
Top-1 match	26 / 30	86.7%
Any@3 match	28 / 30	93.3%
All@3 match	26 / 30	86.7%

#	Expected	Detected top-3 (id : score)	Top-1	Any@3	All@3
1	MSK_001	MSK_001:0.43 PSY_002:0.32 MET_001:0.29	YES	YES	YES
2	GIT_001	GIT_001:0.77 GIT_003:0.62 GIT_008:0.54	YES	YES	YES
11	DERM_002	DERM_002:0.44 NEU_001:0.32 DERM_001:0.25	YES	YES	YES
12	GEN_001	PSY_002:0.43 GEN_001:0.41 GIT_008:0.39	no	YES	YES

BASELINE VERSION 2 - RULE BASED

User Persona

```
{
  "Age": 90,
  "Weight": 92.0,
  "Average_Sleeping_Hours": 6.0,
  "Has_Diabetes": 0,
  "Has_High_Blood_Pressure": 1,
  "Daily_Exercise": 0,
  "Medications": [
    "Lexapro",
    "Tylenol",
    "Tums",
    "Mylanta"
  ],
  "Has_Asthma": -1,
  "Has_Pneumonia_History": -1,
  "Has_Shoulder_Injury": 0,
  "Has_Tremors": 1,
  "Has_Mouth_Sores": 1,
  "Has_Nausea_Vomiting": -1,
  "Has_Stomach_Pain": 1,
  "Has_Heart_Disease": 0
}
```

Pre defined Rules

```
def sleep_blood_pressure(self):
    sleep_hr = self._persona['Average_Sleeping_Hours']
    bp = self._persona['Has_High_Blood_Pressure']
    if sleep_hr != -1 and bp != -1:
        if sleep_hr < 8 and bp == 1:
            self.generate_task("Try to get more sleep to reduce BP")

def exercise_weight(self):
    exercise = self._persona['Daily_Exercise']
    weight = self._persona['Weight']
    if exercise != -1 and weight != -1:
        if exercise != 1 and weight > 90:
            self.generate_task("Try to do more exercise to reduce weight")

def tremors_lexapro(self):
    tremors = self._persona.get('Has_Tremors', -1)
    meds = self._persona.get('Medications', [])
    if tremors == 1 and 'Lexapro' in meds:
        self.generate_task("Monitor tremors as a side effect of Lexapro; consult doctor")
```


ASSESSMENT

Symptoms: Improved right arm Symptoms, subtle tremor.

Diagnosis: Hypertension, depression (improved with Lexapro), osteoporosis, osteoarthritis.

History of Patient: Right arm Symptoms resolved, depression possibly improved with Lexapro (full dosage started about a week ago), occasional non-daily intake of Lexapro, subtle tremor observed.

Plan of Action: Monitor blood pressure, continue Lexapro at full dosage daily, monitor for worsening joint or bone pain related to osteoporosis and osteoarthritis.

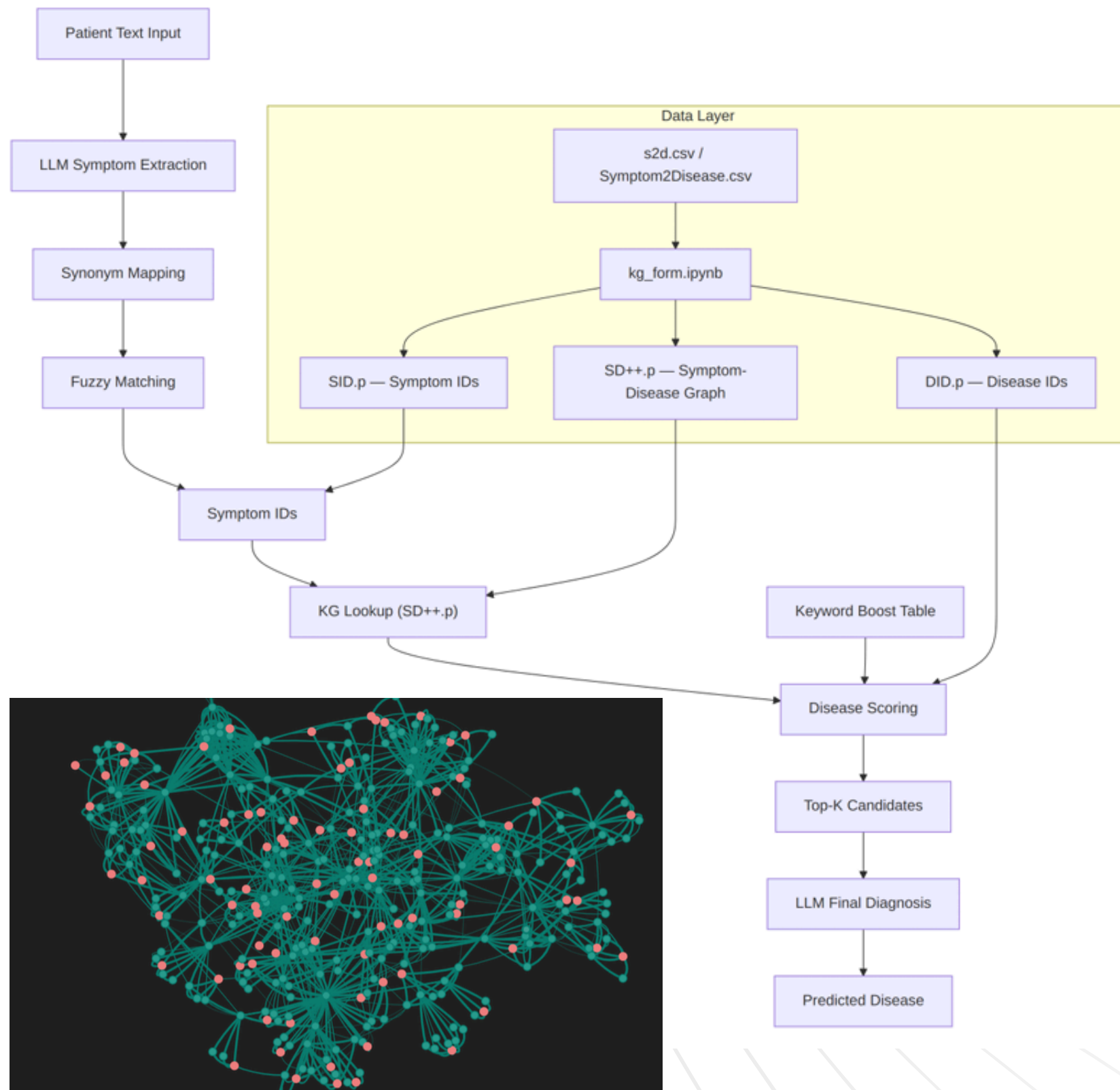
13. Monitor tremors as a side effect of Lexapro; consult doctor if they worsen

14. Monitor blood pressure at home to confirm white coat hypertension

Associated challenges

1. Why you see {... -1 ... Has_High_Blood_Pressure:1 ...} each run
 - write_attr.py loads persona.json, then calls extract_profile using LLM on retrieved conversation.
 - Your output shows only Has_High_Blood_Pressure got updated to 1; others remained -1.
 - This happens if input conversation text only clearly indicates high BP, or LLM extraction wasn't robust on other fields.
2. There has to be a limit to no. of rules that need to be defined

KNOWLEDGE GRAPH-BASED MEDICAL DIAGNOSIS SYSTEM



Problem Statement

Diagnosing diseases from patient-described symptoms is difficult because:

- Patients use non-clinical language
- Symptoms overlap across diseases
- Traditional rule systems are rigid
- LLMs alone may hallucinate or miss structure

Goal

Build an intelligent diagnosis system that:

- Understands natural language symptoms
- Maps them to medical terminology
- Predicts the most likely diseases accurately

Objectives

- Extract symptoms from free-text patient descriptions
- Build a Knowledge Graph connecting symptoms and diseases
- Use LLMs for:
 - Symptom extraction
 - Final disease ranking
- Improve accuracy using:
 - Synonym mapping
 - Fuzzy matching

Knowledge Graph Construction Weighting Method

TF-IDF inspired weighting:

- Rare symptoms get higher importance
- Common symptoms get lower importance

System Architecture

Pipeline Flow

1. Patient enters symptoms in natural language
2. LLM extracts symptoms
3. Synonym & fuzzy matching standardize terms
4. Symptoms mapped to KG IDs
5. Knowledge Graph scoring performed
6. LLM ranks final disease candidates

Symptom Extraction

Features

- Extracts only explicit symptoms
- Splits combined phrases
- Maintains medical terminology

Symptom Matching Cascade

Stage 1 — Exact Match

- "headache" → "headache"

Stage 2 — Synonym Mapping

- "chest pain" → "sharp chest pain"
- "confusion" → "disturbance of memory"

Stage 3 — Fuzzy Matching

- "breathlessness" → "shortness of breath"

Disease Scoring Algorithm

Disease Scoring

Step 1 — KG Scoring

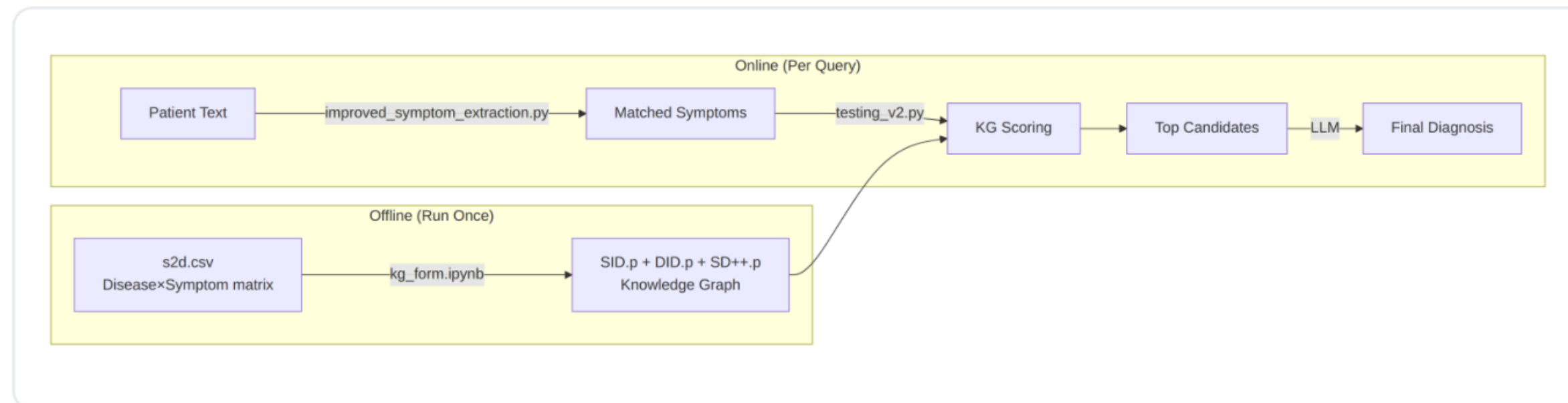
Aggregate symptom-disease edge weights

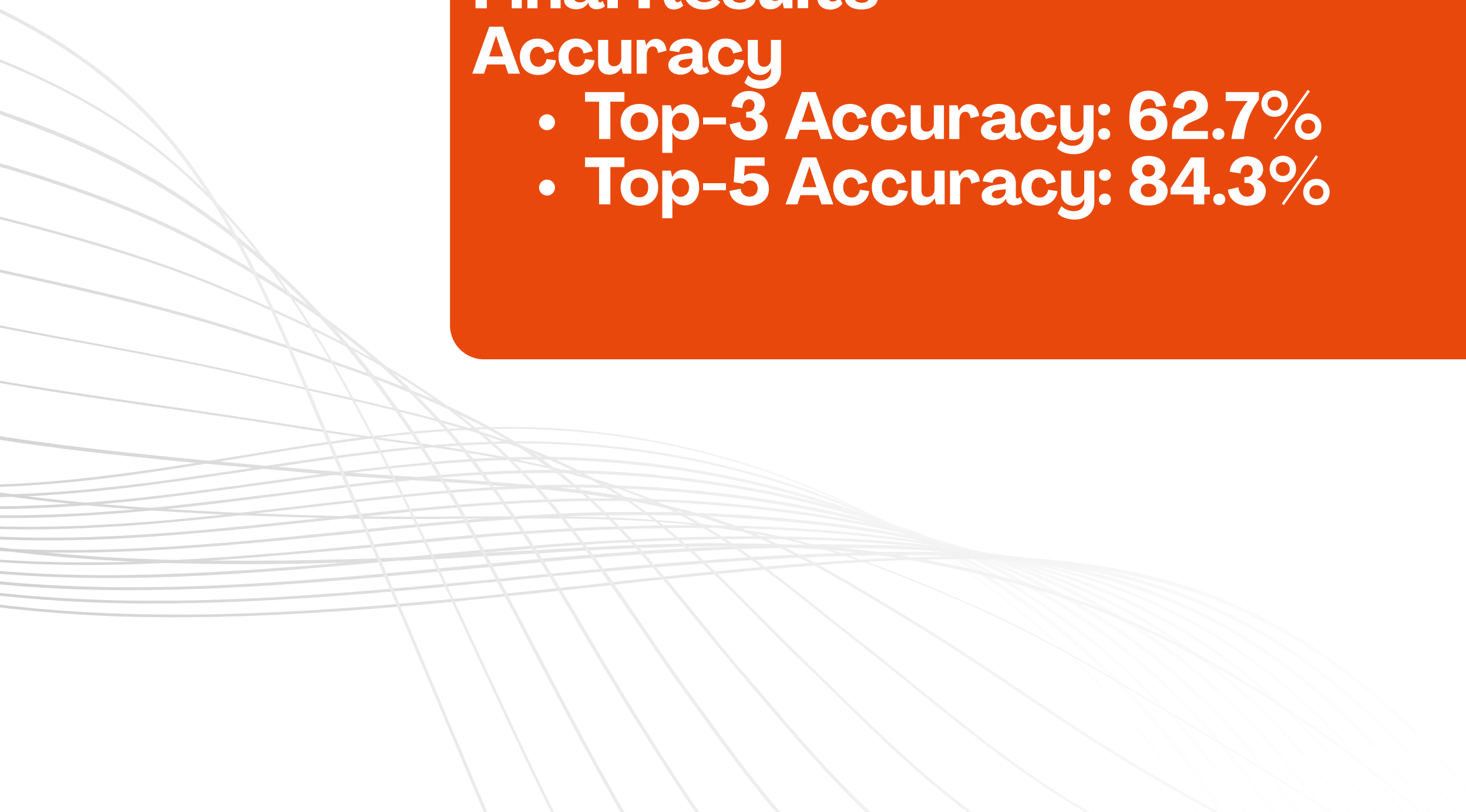
Step 2 — Coverage Calculation

Measure how many symptoms match each disease

Final Score Formula :

$$0.50 * \text{Avg_Score} + 0.50 * \text{Coverage}$$

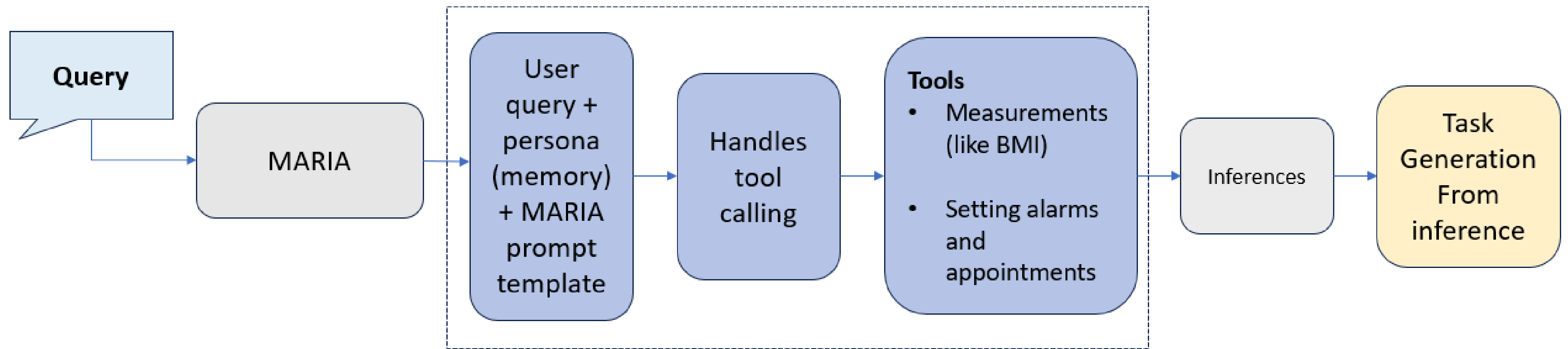




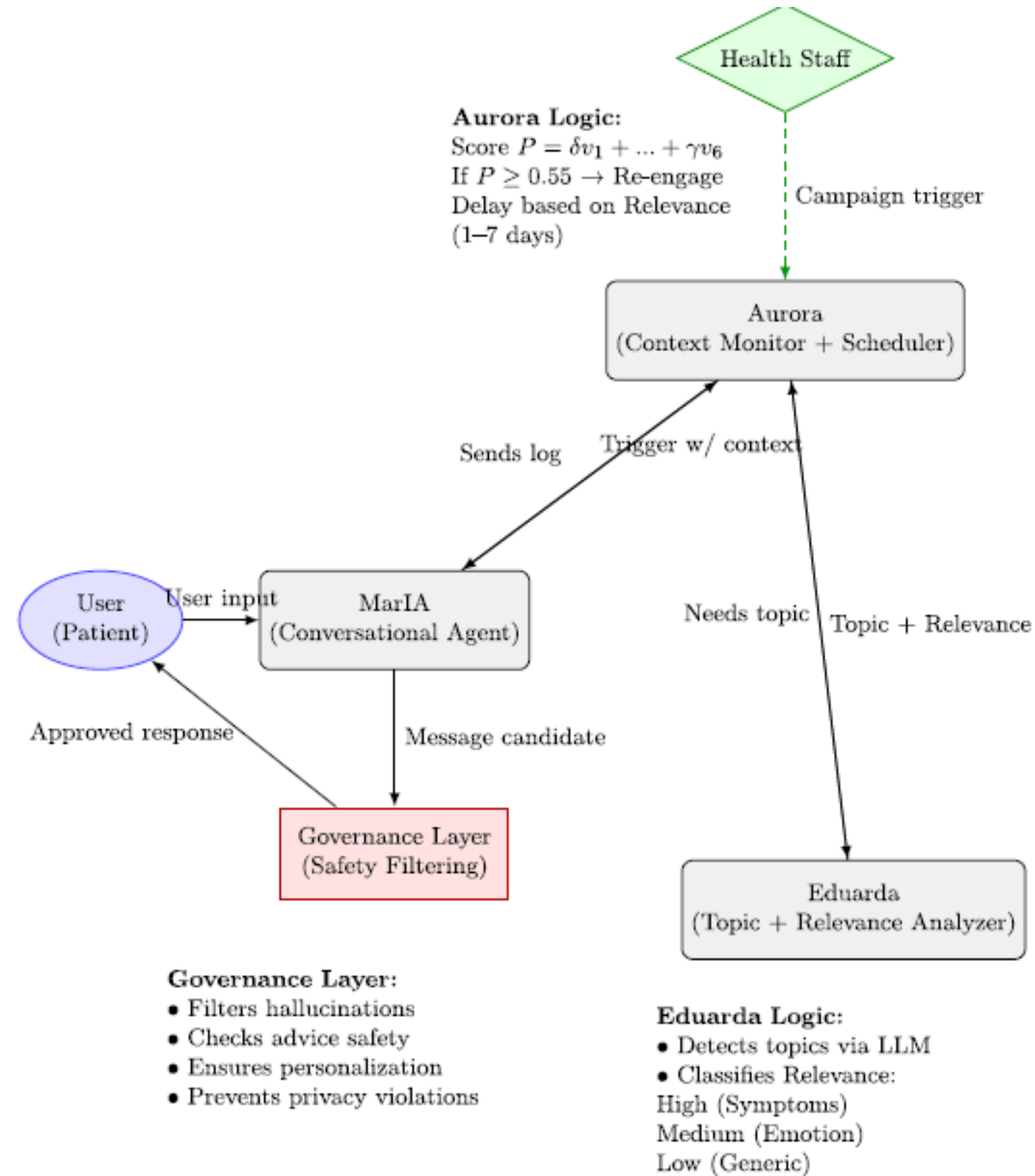
Final Results Accuracy

- **Top-3 Accuracy: 62.7%**
- **Top-5 Accuracy: 84.3%**

MARIA Integration in our pipeline



MARIA framework

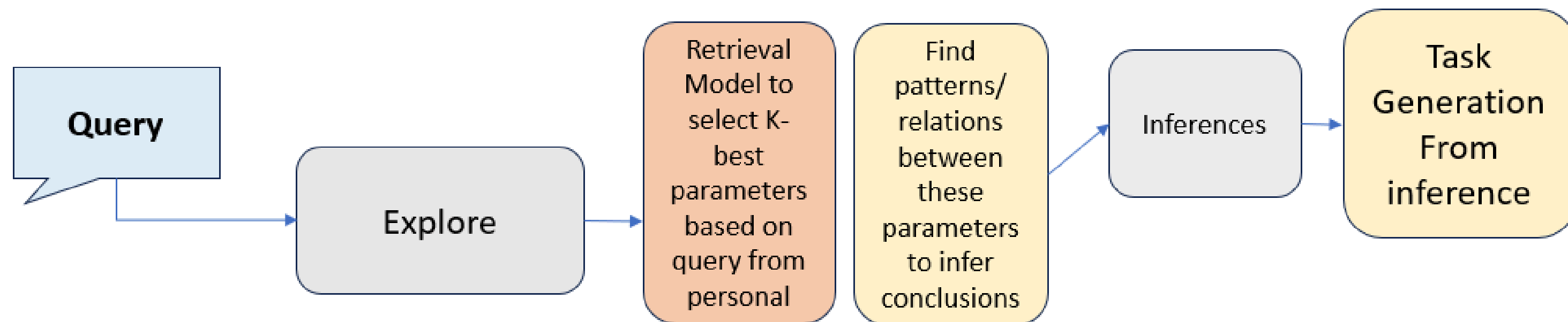


Exploration

Based on user query, select top-K persona parameters and track their change in values over a time period (delta)

Try to find relations between these parameters and get inference → Use this to generate tasks

- In our current setup and experiments, the retrieval is handled by LLM
- Further steps include training our own retrieval model



Full Pipeline - Results

Rule based + MARIA + KG + LLM Exploration

- We ask an independent LLM model to rank the response (tasks generated) against the user query on Likert Scale (0-5) based on how suitable it is for the given query and how specific it is to the particular user query.
- We average the Likert Score (out of 5) over 30 responses

Dataset

We created a list of 30 related conversations for a sample user and evaluated the response of the LLM against these queries

Judge Model Name	Likert Score (Our Model)	Likert Score (LLM Only)	Likert Score (LLM+CoT)
GPT OSS 120b	4.08	3.15	3.38
Groq-compound	3.88	3.08	3.22